

CONTACT INFORMATION	School of Computing University of Nebraska-Lincoln Office: 262 Avery Hall, Lincoln, NE, United States Email: qiang.liu@unl.edu Tel: 402-472-5006 Website: https://cse.unl.edu/~qliu/		
RESEARCH INTERESTS	Wireless Network, Edge Computing, Machine Learning, Cyber-Physical System		
EDUCATION	The University of North Carolina at Charlotte , Charlotte, NC		2016–2020
	• Ph.D. in Electrical Engineering		
	University of Electronic Science and Technology of China , Chengdu, China		2013–2016
	• M.S. in Communication and Information System		
	Jiangxi Normal University , Nanchang, China		2009–2013
	• B.S. in Electronics and Information Engineering		
HONORS AND AWARDS	◇ <i>Academic Grant Program Award</i> , NVIDIA		2025
	◇ <i>Edgerton Innovation Award</i> , College of Engineering, UNL		2025
	◇ <i>Research Excellence Award</i> , College of Engineering, UNL		2024
	◇ <i>Best Paper Award</i> , IEEE International Conference on Communications (ICC)		2022
	◇ <i>Best Presentation Award</i> , ACM CoNEXT		2021
	◇ <i>Best Paper Award</i> , IEEE ComSoc on Transmission, Access, and Optical Systems (TAOS)		2019
	◇ <i>Best Paper Award</i> , IEEE International Conference on Communications (ICC)		2019
	◇ <i>Outstanding Graduate Student Award</i> , UNC-Charlotte		2019
GRANTS	<i>In total, 18 grants (\$6.14M in total) from multiple federal and state agencies, internal and industry, with PI amount 3.42M (my portion \$2.60M) and Co-PI amount 2.72M (my portion \$545K).</i>		
	• National Science Foundation, CAREER: Trustworthy AI-Native Network Autonomy in Open Radio Access Networks, Sole PI, \$750,000 2026 - 2031 • Nebraska Collaboration Initiative (NRI), NARS: Networked Autonomous Robotic Systems for Advanced Manufacturing, PI, \$99,550 2026 - 2027 • UNL Fred J. Kelly Fund, AI-Teach-to-Learn and Teach-AI-to-Learn for Computer Science Courses, PI, \$99,576 2026 - 2027 • Department of Energy-NREL, Modernizing EV Emergency Response Training Through XR and AI for Greater Engagement and Reach, Co-PI, \$19,500 (my portion ~\$5K) 2025 - 2026 • National Science Foundation, Collaborative Research: CSR: Small: CADaaS: Deterministic Communication and Predictive Computation for Connected Autonomous Driving as a Service, Lead PI, \$289,374 2025 - 2028 • Nebraska DoT, Evaluating the Interoperability of Connected and Autonomous Vehicles and Signal Phasing and Timing Infrastructure, Co-PI, \$185,796, (my portion ~\$48K) 2025 -2027 • Nebraska DoT, Development of Reliable Wireless Connectivity in Support of e-Ticketing System in Nebraska Rural Construction Sites, Co-PI, \$173,485, (my portion ~\$55K) 2025 -2027 • National Science Foundation, EPSCoR Research Fellows: NSF: Online Hierarchical Learning for Network Autonomy in Open Radio Access Networks, Sole PI, \$285,694 2025 - 2026 • US DoT FMCSA, Roadside-to-Vehicle Crash Avoidance Warning System for Commercial Motor Vehicles on Rural Roads, Co-PI, \$1,342,761 (my portion ~\$200K) 2024 -2026		

- **USDA NIFA**, CropTwin: Automatic Digital Twin for Crop Growth Modeling towards Smart Irrigation Management, **PI**, \$300,000 2024 -2026
- **National Science Foundation**, NeTS: Small: AutoSlicing: Safe Online Autonomous Network Orchestration Towards Pervasive Slicing-as-a-Service, **Sole PI**, \$590,339 2024 - 2027
- **NASA Nebraska Space Grant**, Bring Advanced Cellular Networks From Earth to Space, **Sole PI**, \$5,000 2024 - 2025
- **Toyota Motor North America**, Serverless Edge Computing for Large Machine Learning Models: Fundamentals, Optimization, and Use Cases, **Sole PI**, \$50,000 2024 - 2025
- **National Science Foundation**, CC* Integration-Large: Husker-Net: Open Nebraska End-to-End Wireless Edge Networks, **PI**, \$891,000 2023-2025
- **Nebraska EPSCoR FIRST Award**, Real-World Machine Learning in Mobile Network Slicing, **Sole PI**, \$25,000 2023 - 2024
- **National Science Foundation**, CNS Core: Medium: Field-Nets: Field-to-Edge Connectivity for Joint Communication and Sensing in Next-Generation Intelligent Agricultural Networks, **Co-PI**, \$1,000,000 (my portion \$157K) 2022 - 2025
- **Toyota Motor North America**, Collaborative Data Offloading and Resource Provisioning for Crowdsourcing HD Map in Automotive Edge Computing, **Sole PI**, \$50,000 2022 - 2023
- **UNL Layman Fund**, Automated Offline Simulator Augmentation with Real-to-Sim Learning in Mobile Networks, **Sole PI**, \$10,000 2022 - 2023

PUBLICATIONS **Conferences**

34. Y. Zhang, M. Zhao, Q. Liu, A. Alkhateeb, A. K. Agrawal, and Q. Qu, “RadTwin: Generalizable Wireless Digital Twin for Dynamic Environments”, *International Conference on Computer Communications and Network (ICCCN)*, Hawaii, USA. May. 2026
33. J. Cao, X. Li, Q. Liu, T. Han, N. Zhang, and W. Shi, “Edgeserving: Deadline-aware multi-DNN serving at the edge”, *International Conference on Computer Communications and Network (ICCCN)*, Hawaii, USA. May. 2026
32. X. Li, Y. Zhang, Q. Liu, M. C. Vuran, N. Huynh, L. Zhao, M. Rahman, and E. E. Ozguven, “Automatic network planning with digital radio twin,” in *ICC 2026-IEEE International Conference on Communications*, *IEEE International Conference on Communications (ICC)*, Scotland, UK. May. 2026
31. M. Zhao, Y. Zhang, Q. Liu, A. Kak, N. Choi, “inRAN: Interpretable Online Bayesian Learning for Network Automation in Open Radio Access Networks”, *IEEE International Conference on Computer Communications (INFOCOM)*, Tokyo, Japan. May. 2026
30. Y. Zhang, M. Zhao, Q. Liu, N. Choi, “oneTwin: Online Digital Network Twin via Neural Radio Radiance Field”, *IEEE International Conference on Computer Communications (INFOCOM)*, Tokyo, Japan. May. 2026
29. M. Zhao, Y. Zhang, Q. Liu, A. Kak, N. Choi, “InSlicing: Interpretable Learning-Assisted Network Slice Configuration in Open Radio Access Networks”, *IEEE International Conference on Computer Communications (INFOCOM) NG-OPERA*, London, United Kingdom. May. 2025
28. M. Zhao, Y. Zhang, Q. Liu, A. Kak, N. Choi, “AdaSlicing: Adaptive Online Network Slicing under Continual Network Dynamics in Open Radio Access Networks”, *IEEE International Conference on Computer Communications (INFOCOM)*, London, United Kingdom. May. 2025
27. M. Zhao, Y. Zhang, Q. Liu, T. Han, “LeFi: Learn to Incentivize Federated Learning in Automotive Edge Computing”, *IEEE Global Communications Conference (GLOBECOM)*, Cape Town, South Africa. Dec. 2024
26. J. Cao, Q. Liu, D. Chen, and K. Han, “CAVE: Crowdsourcing Passing-By Vehicles for Reliable In-Vehicle Edge Computing”, *IEEE Global Communications Conference (GLOBECOM)*, Cape Town, South Africa. Dec. 2024
25. Y. Zhang, M. Zhao, Q. Liu, N. Choi, “Learn to Augment Network Simulators Towards Digital Network Twins”, *IEEE INFOCOM Workshop (NG-OPERA)*, Vancouver, Canada. May. 2024

24. Q. Liu, Y. Xue, Y. Zhang, D. Chen, and K. Han, “AdaMap: High-Scalable Real-Time Cooperative Perception at the Edge”, *IEEE/ACM Symposium on Edge Computing (SEC)*, Wilmington, DE, Dec. 2023 (acceptance rate: **25.4%**)
23. T. Hu, Q. Liao, Q. Liu, A. Massaro, G. Carle, “Fast and Scalable Network Slicing by Integrating Deep Learning with Lagrangian Methods”, *IEEE Global Communications Conference (GLOBECOM)*, Kuala Lumpur, Malaysia. 2023
22. Y. Xue, Y. Zhang, Q. Liu, D. Chen, and K. Han, “CoMap: Proactive Provision for Crowdsourcing Map in Automotive Edge Computing”, *IEEE International Conference on Communications (ICC)*, Rome, Italy, June. 2023
21. Y. Zhang, Y. Xue, Q. Liu, N. Choi, and T. Han, “RoNet: Toward Robust Neural Assisted Mobile Network Configuration”, *IEEE International Conference on Communications (ICC)*, Rome, Italy, June. 2023
20. Q. Liu, N. Choi, T. Han, “Atlas: Automate Online Service Configuration in Network Slicing”, *The 18th International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, Rome, Italy, Dec. 2022 (acceptance rate: **18.5%**)
19. Q. Liu, Y. Zhang, H. Wang, “EdgeMap: CrowdSourcing High Definition Map in Automotive Edge Computing”, *IEEE International Conference on Communications (ICC)*, Virtual, May. 2022
18. T. Hu, Q. Liao, Q. Liu, G. Carle, “Network Slicing via Transfer Learning aided Distributed Deep Reinforcement Learning”, *IEEE Global Communications Conference (GLOBECOM)*, Rio de Janeiro, Brazil, Dec. 2022
17. T. Hu, Q. Liao, Q. Liu, D. Wellington, G. Carle, “Inter-Cell Slicing Resource Partitioning via Coordinated Multi-Agent Deep Reinforcement Learning”, *IEEE International Conference on Communications (ICC)*, Virtual, May. 2022 (**Best Paper Award**)
16. Q. Liu, N. Choi, T. Han, “OnSlicing: Online End-to-End Network Slicing with Reinforcement Learning”, *The 17th International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, Virtual, Dec. 2021 (acceptance rate: **22%**)
15. Q. Liu, N. Choi, T. Han, “Constraint-Aware Deep Reinforcement Learning for End-to-End Resource Orchestration in Mobile Networks”, *IEEE International Conference on Network Protocols (ICNP)*, Virtual, Nov. 2021 (acceptance rate: **24%**)
14. Q. Liu, T. Han, L. Xie, B. Kim, “LiveMap: Real-Time Dynamic Map in Automotive Edge Computing”, *IEEE International Conference on Computer Communications (INFOCOM)*, Virtual, May 2021 (acceptance rate: **19.9%**)
13. Q. Liu, T. Han, E. Moges, “EdgeSlice: Slicing Wireless Edge Computing Network with Decentralized Deep Reinforcement Learning”, *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Singapore, Dec. 2020 (acceptance rate: **18%**)
12. Q. Liu, T. Han, N. Zhang, Y. Wang, “DeepSlicing: Deep Reinforcement Learning Assisted Resource Allocation for Network Slicing”, *IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, Dec. 2020
11. Q. Liu, T. Han, “DIRECT: Distributed Cross-Domain Resource Orchestration in Cellular Edge Computing”, *ACM International Symposium on Mobile Ad Hoc Networking and Computing (MOBIHOC)*, Catania, Italy, Jul. 2019 (acceptance rate: **23.7%**)
10. Q. Liu, T. Han, “VirtualEdge: Multi-Domain Resource Orchestration and Virtualization in Cellular Edge Computing”, *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Dallas, TX, Jul. 2019 (acceptance rate: **19.6%**)
9. Q. Liu, T. Han, “DARE: Dynamic Adaptive Mobile Augmented Reality with Edge Computing”, *IEEE International Conference on Network Protocols (ICNP)*, Cambridge, UK, Sep. 2018 (acceptance rate: **17.8%**)
8. Q. Liu, S. Huang, J. Opadere, T. Han, “An Edge Network Orchestrator for Mobile Augmented Reality”, *IEEE International Conference on Computer Communications (INFOCOM)*, Honolulu, HI, Apr. 2018 (acceptance rate: **19.2%**)

7. J. Opadere, Q. Liu, N. Zhang, T. Han, "Joint Computation and Communication Resource Allocation for Energy-Efficient Mobile Edge Networks", *IEEE International Conference on Communications (ICC)*, Shanghai, China, May 2019 (**Best Paper Award**)
6. Q. Liu, T. Han, "Energy-Efficient On-demand Cloud Radio Access Networks Virtualization", *IEEE Global Communications Conference (GLOBECOM)*, Abu Dhabi, UAE, Dec. 2018
5. Q. Liu, T. Han, N. Ansari, "Joint Radio and Computation Resource Management for Low Latency Mobile Edge Computing", *IEEE Global Communications Conference (GLOBECOM)*, Abu Dhabi, UAE, Dec. 2018
4. J. Opadere, Q. Liu, T. Han, "Energy-Efficient RRH Sleep Mode for Virtual Radio Access Networks", *IEEE Global Communications Conference (GLOBECOM)*, Singapore, Dec. 2017
3. Q. Liu, G. Wu, Y. Guo, Y. Zhang, S. Hu, "Energy Efficient Resource Allocation for Control Data Separated Heterogeneous-CRAN", *IEEE Global Communications Conference (GLOBECOM)*, Washington DC, Dec. 2016
2. Q. Liu, T. Han, G. Wu, "Computing Resource Aware Energy Saving Scheme for Cloud Radio Access Networks", *IEEE Sustainable Computing and Communications (SustainCom)*, Atlanta, GA, Oct. 2016
1. Y. Guo, Q. Liu, G. Wu, S. Li, "On the Impact of Power Amplifier Efficiency on the Energy Efficiency in a Massive MIMO System", *WiCOM*, Shanghai, China, 2015

Journal and Magazines

12. T. Hu, Q. Liao, Q. Liu, G. Carle, "Information Bottleneck-Based Domain Adaptation for Hybrid Deep Learning in Scalable Network Slicing", *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 2, pp.1642 - 1660, Oct. 2024
11. Y. Zhang, Q. Liu, H. Wang, D. Chen, and K. Han, "CrowdSourcing Live High Definition Map via Collaborative Computation in Automotive Edge Computing", *IEEE Transactions on Vehicular Technology*, 2024
10. T. Hu, Q. Liao, Q. Liu, G. Carle, "Inter-Cell Network Slicing with Transfer Learning Empowered Multi-Agent Deep Reinforcement Learning", *IEEE Open Journal of the Communications Society*, vol. 4, pp.1141 - 1155, May 2023
9. H. Wang, Z. Wang, D. Chen, Q. Liu, H. Ke, K. Han, "Metamobility: Connecting Future Mobility With the Metaverse", *IEEE Vehicular Technology Magazine*, 2023
8. Q. Liu, N. Choi, T. Han, "Deep Reinforcement Learning for End-to-End Network Slicing: Challenges and Solutions", *IEEE Network Magazine*, 2022
7. Q. Liu, T. Han, J. Xie, and B. Kim, "Real-Time Dynamic Map with Crowdsourcing Vehicles in Edge Computing", *IEEE Transactions on Intelligent Vehicles*, pp.1-10, 2022
6. F. Salahdine, Q. Liu, T. Han, "Towards Secure and Intelligent Network Slicing for 5G Networks", *IEEE Open Journal of the Computer Society*, vol. 3, pp.23-38, Mar 2022
5. F. Salahdine, J. Opadere, Q. Liu, T. Han, N. Zhang, S. Wu, "A survey on sleep mode techniques for ultra-dense networks in 5G and beyond", *Computer Networks*, vol. 201, pp.108567, 2021
4. Q. Liu, T. Han, N. Ansari, "Learning-Assisted Secure End-to-End Network Slicing for Cyber-Physical Systems", *IEEE Network Magazine*, vol. 34, no. 3, pp. 37-43, May 2020
3. J. Opadere, Q. Liu, T. Han, N. Ansari, "Energy-efficient Virtual Radio Access Networks for Multi-Operators Cooperative Cellular Networks", *IEEE Transactions on Green Communications and Networking (TGCN)*, vol. 3, no. 3, pp. 603-614, Sep. 2019
2. Q. Liu, T. Han, N. Ansari, "Energy-Efficient On-demand Resource Provisioning in Cloud Radio Access Networks", *IEEE Transactions on Green Communications and Networking (TGCN)*, vol. 3, no. 4, pp. 1142-1151, Jul. 2019
1. Q. Liu, T. Han, N. Ansari, G. Wu, "On Designing Energy-Efficient Heterogeneous Cloud Radio Access Networks", *IEEE Transactions on Green Communications and Networking (TGCN)*, vol. 2, no. 3, pp. 721-734, May 2018

Posters and Demos

5. Y. Zhang, Y. Xue, Q. Liu, N. Choi, “Poster: Digital Network Twin via Learning-Based Simulator”, *IEEE International Conference on Computer Communications (INFOCOM)*, Hoboken, NJ, May. 2023
4. Q. Liu, T. Han, “When Network Slicing meets Deep Reinforcement Learning”, *ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT)* Student Workshop, Orlando, FL, Dec. 2019
3. Q. Liu, T. Han, “Demo Abstract: Themis: Cross-Domain Resource Orchestration and Virtualization in Cellular Computing Networks”, *IEEE International Conference on Network Protocols (ICNP)*, Cambridge, UK, Sep. 2018
2. Q. Liu, S. Huang, T. Han, “Demo Abstract: Fast and Accurate Object Analysis at the Edge for Mobile Augmented Reality”, *ACM/IEEE Symposium on Edge Computing (SEC)*, San Jose, CA, Oct. 2017
1. Q. Liu, S. Huang, Y. Deng, T. Han, “Demo Abstract: MExR: Mobile Edge Resource Management for Mixed Reality Applications”, *IEEE International Conference on Computer Communications (INFOCOM)*, Atlanta, GA, Apr. 2017

ACADEMIC EXPERIENCE	• University of Nebraska-Lincoln Assistant Professor	Aug. 2021–Present
	• University of North Carolina at Charlotte Research and Teaching Assistant	Aug. 2016–Dec. 2020
INDUSTRY EXPERIENCE	• Nokia Bell Labs Member of Technical Staff	Jan. 2021–Aug. 2021
	• Nokia Bell Labs Research Intern	Jun. 2020–Aug. 2020
	• Toyota InfoTech Labs Research Intern	Jan. 2020–Jun. 2020
	• Meta Reality Labs Research Intern	May. 2019–Nov. 2019
TEACHING	◇ Instructor, CSCE 990: Multi-Access Edge Computing, Fall 2026, UNL ◇ Instructor, CSCE 464/864: Internet System and Programming, Spring 2026, UNL ◇ Instructor, CSCE 990: Multi-Access Edge Computing, Fall 2025, UNL ◇ Instructor, CSCE 464/864: Internet System and Programming, Spring 2025, UNL ◇ Instructor, CSCE 990: Multi-Access Edge Computing, Fall 2024, UNL ◇ Instructor, CSCE 464/864: Internet System and Programming, Spring 2024, UNL ◇ Instructor, CSCE 990: Multi-Access Edge Computing, Fall 2023, UNL ◇ Instructor, CSCE 464/864: Internet System and Programming, Spring 2023, UNL ◇ Instructor, CSCE 990: Multi-Access Edge Computing, Fall 2022, UNL ◇ Instructor, CSCE 464/864: Internet System and Programming, Spring 2022, UNL ◇ Instructor, CSCE 990: Multi-Access Edge Computing, Fall 2021, UNL ◇ Teaching Assistant, Power Electronics I, Fall 2018, UNCC ◇ Teaching Assistant, Computer Utilization in C++, Spring 2018, UNCC ◇ Teaching Assistant, Data Communications and Networking, Spring 2018, UNCC ◇ Teaching Assistant, Signals and Systems II, Fall 2017, UNCC ◇ Teaching Assistant, Logic and Networks, Spring 2017, UNCC ◇ Teaching Assistant, Signals and Systems I, Fall 2016, UNCC	

SERVICE

- **Editor**, IEEE Wireless Communication Magazine
- **Associate Editor**, Multimedia Tools and Applications, Track 7: Connected and Autonomous Vehicles
- **TPC Co-Chairs**, IEEE INFOCOM WKSHPS: NG-OPERA 2023, 2024, 2025, 2026
- **TPC Member**, IEEE INFOCOM, IEEE ICDCS, IEEE ICCCN, IEEE ICC, IEEE/ACM SEC, ACM 5G-MeMU Workshop
- **Poster Co-Chairs**, The Seventh ACM/IEEE Symposium on Edge Computing (SEC) 2022, 2023
- **Guest Editor**, MDPI Electronics Special Issue "Machine Learning for Next-Generation Wireless Networks and Computing Systems" 2022
- **Reviewer**, IEEE ToN, IEEE TWC, IEEE TMC, IEEE OJCS, IEEE JSAC, IEEE TCCN, IEEE Access, IEEE TGCN, IEEE Communication Letters, IEEE ICC, IEEE GLOBECOM, IEEE System Journal, Elsevier Measurement, Elsevier Computer Communications, Elsevier Computer Networks, Digital Communication and Networks